quality level of the manufactured product. This is done by 'Process Control'. In Process Control the proportion of defective items in the production process is to be minimized and it is achieved through the technique of control charts. Product Control means that controlling the quality of the product by critical examination through sampling inspection plans. Product Control aims at a certain quality level to be guaranteed to the customers. It attempts to ensure that the product sold does not contain a large number of defective items. Thus it is concerned with classification of raw materials, semi-finished goods or finished goods into acceptable or rejectable products.

Control Charts

In an industry, there are two kinds of problems to be faced, namely

(i) To check whether the process is conforming to its standard level.

(ii) To improve the standard level and reduce the variability.

Shewhart's control charts provide an answer to both. It is a simple technique used for detecting patterns of variations in the data. Control charts are simple to construct and easy to interpret. A typical control charts consists of the following three lines.

(i) Centre Line (CL) indicates the desired standard level of the process.

(ii) Upper Control Limit (UCL) indicates the upper limit of tolerance.

(iii) Lower Control Limit (LCL) indicates the lower limit of tolerance.

If the data points fall within the control limits, then we can say that the process is in control, instead if one or more data points fall outside the control limits, then we can say that the process is out of control.

For example, the following diagram shows all the three control lines with the data points plotted, since all the points falls within the control limits, we can say that the process is in control.

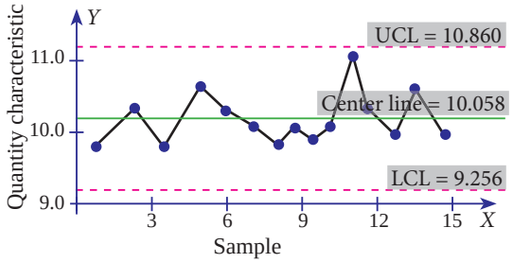


Fig 9.4

Control Charts for Variables

These charts may be applied to any quality characteristic that can be measured quantitatively. A quality characteristic which can be expressed in terms of a numerical value is called as a variable. Many quality characteristics such as dimensions like length, width, temperature, tensile strength etc… of a product are measurable and are expressed in a specific unit of measurements. The variables are of continuous type and are regarded to follow normal probability law. For quality control of such data, there are two types of control charts used. They are as follows :

(i) Charts for Mean ($\overline{X}$)

(ii) Charts for Range (R)

9.3.4 Construction of $\overline{X}$ and R charts

Any production process is not perfect enough to produce all the products exactly the same. Some amount of variation is inherent in any production process. This variation is a total of number of characteristics of the production process such as raw materials, machine setting, operators, handling new operations and new machines, etc. The $\overline{X}$ chart is used to show the quality averages of the samples taken from the given process. The *R* chart is used to show the

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